Expected Payoff

Name _____

Is a lottery ticket worth buying? Is a protection plan worth the money? The expected value of the **payoff** tells you what happens on average.

Net gain = what you win − what you paid
Expected payoff = Σ (net gain $\cdot$ probability)

Example 1: a lottery ticket. A ticket costs \$2. 1 in 1,000 tickets wins \$500. 1 in 50 tickets wins \$10. The rest win nothing.

Outcome	Net gain x	$P(x)$	$x \cdot P(x)$
Win \$500	$500 - 2 = 498$	0.001	0.498
Win \$10	$10 - 2 = 8$	0.02	0.16
Win nothing	-2	0.979	-1.958
	$E(X) = \Sigma\, x \cdot P(x)$		-1.30

On average, a buyer **loses \$1.30** per ticket. The lottery keeps that money.

Example 2: a protection plan. A phone costs \$300 to replace. The chance it breaks this year is 0.10. A protection plan costs \$40 for the year.

Expected cost with no plan: $0.10 \cdot 300 = \$30$
Cost of the plan: \$40

On average, the plan costs \$10 more than paying for a broken phone yourself. That is how the company makes money.

But: the plan protects you from a sudden \$300 bill. Expected value is one part of a decision. How much risk you can afford is another.

Work with a partner. Compare answers when you finish each problem.

1. A school sells 500 raffle tickets for $5 each. One ticket wins $1,000.

   (a) Fill in the table.

Outcome	Net gain x	$P(x)$	$x \cdot P(x)$
Win			
Lose			

   (b) Find the expected payoff. What does it mean?

2. A game costs $1 to play. You spin a spinner: half the time you win nothing, a quarter of the time you win $1, and a quarter of the time you win $3.

   (a) Find the expected payoff.

   (b) Is the game fair?

3. At a carnival, you pay $2 and draw one card from a deck. An ace wins $10. A face card (J, Q, K) wins $5. Anything else wins nothing.

   (a) Find the probability of each outcome.

   (b) Find the expected payoff. Should you expect to win money?

4. A laptop repair costs $400. The chance a laptop needs repair in its first year is 0.15. A one-year warranty costs $75.

   (a) Find the expected repair cost without the warranty.

   (b) Which is cheaper on average: buying the warranty or not?

   (c) Give a reason someone might still buy the warranty.

5. The store sells 1,000 of these warranties. About how much money does the store expect to make? Explain.